Amazon EBS — 30 Important Interview Questions & Answers

1. What is Amazon EBS?
Answer:
Amazon EBS stands for **Amazon Elastic Block Store**. It is block storage designed mainly for EC2 instances. We can use EBS volumes to store operating systems, applications, databases, and other data.
For example:
When we launch an EC2 instance, we normally attach an EBS volume to store the operating system and application files.
Example:
EC2
↓
EBS Volume
↓
OS + Application + Data
Simple ga gurthupettuko:
EBS = Block Storage for EC2

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2. Why do we use EBS?
Answer:
EBS provides persistent block storage for EC2 instances. Data stored on EBS can remain available even when the EC2 instance is stopped. We can also create snapshots, change volume size, and select different volume types based on performance requirements.
For example:
If an EC2 server is stopped and started again, the data on its EBS volume can still be available.

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3. What is an EBS Volume?
Answer:
An EBS Volume is a virtual block-storage device that can be attached to an EC2 instance. It can store operating-system files, application files, databases, and other data.
For example:
EC2
↓
EBS Volume
↓
/data
↓
Application Files

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4. What are the main EBS Volume Types?
Answer:
EBS provides different volume types for different workloads. Common types include general-purpose SSD volumes, provisioned-IOPS SSD volumes, and HDD-based volumes for throughput-focused workloads. We choose the volume type based on performance, cost, and workload requirements.
For example:
General Application → General Purpose SSD

High IOPS Database → Provisioned IOPS SSD

Large Throughput Workload → HDD-based option

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5. What is gp3?
Answer:
gp3 is a general-purpose SSD EBS volume type. It allows us to configure storage, IOPS, and throughput independently within supported limits. It is commonly used for many general-purpose applications.
For example:
A normal web application running on EC2 can use a gp3 EBS volume for its operating system and application data.

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6. What is io2?
Answer:
io2 is a high-performance EBS volume type designed for workloads that need high and consistent IOPS with strong durability requirements. It is commonly considered for demanding database workloads.
For example:
A critical database application that needs high I/O performance can use an appropriate io2 configuration.

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7. What is an EBS Snapshot?
Answer:
An EBS Snapshot is a point-in-time backup of an EBS volume. Snapshots are stored in Amazon S3-managed infrastructure and can be used to create new EBS volumes or restore data.
For example:
EBS Volume
↓

Snapshot
↓
Create New EBS Volume

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8. Why do we create EBS Snapshots?
Answer:
We create snapshots to protect important data and create backups before risky changes. Snapshots can also be used when we need to create another volume from an existing volume.
For example:
Before upgrading an application, we can create an EBS snapshot so that we have a backup point.

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9. What is an EBS Snapshot Archive?
Answer:
EBS Snapshot Archive is a lower-cost storage option for snapshots that we do not need to access frequently. It is useful for long-term backup and archive requirements where reducing storage cost is more important than quick access.
For example:
Old compliance backups that are rarely needed can be moved to snapshot archive storage.

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10. What is EBS Encryption?
Answer:
EBS encryption protects data stored on EBS volumes and snapshots. AWS KMS keys can be used to manage encryption keys. Encryption helps protect sensitive application and database data.
For example:
EC2
↓
Encrypted EBS Volume
↓
Encrypted Data

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11. What is an encrypted EBS Volume?
Answer:
An encrypted EBS volume stores data in encrypted form. Encryption also applies to supported EBS snapshots and data moving between supported EBS infrastructure components. It is recommended for sensitive workloads.
For example:
A banking application can use encrypted EBS volumes to protect customer-related data.

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12. Can we increase EBS Volume size?
Answer:
Yes. EBS volumes can be modified to increase their size and, depending on the volume type, adjust performance settings such as IOPS or throughput. The operating system may also need additional steps to use the newly added space.
For example:
100 GB
↓
Modify Volume
↓
200 GB

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13. What is EBS Elastic Volumes?
Answer:
Elastic Volumes allow us to modify supported EBS volume properties without replacing the volume. We can increase storage and change supported performance settings while the volume is attached, depending on the workload and volume type.
For example:
Existing EBS
↓
Modify
↓
More Storage / Performance

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14. What is IOPS?
Answer:
IOPS means **Input/Output Operations Per Second**. It represents how many input and output operations a storage volume can handle per second. Higher IOPS is useful for workloads that perform many small read and write operations.
For example:
A busy database may need high IOPS because it performs many read and write operations.

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15. What is EBS Throughput?
Answer:
Throughput is the amount of data that can be transferred between the storage volume and the instance over time. It is usually measured in MB/s. Throughput is important for workloads that process large amounts of sequential data.
For example:
A workload processing large files may care more about throughput than extremely high IOPS.

16. What is the difference between IOPS and Throughput?
Answer:
IOPS tells us how many I/O operations the storage can handle per second. Throughput tells us how much data can be transferred per second. Database workloads often care about IOPS, while large file-processing workloads may care more about throughput.
For example:
IOPS → Number of operations

Throughput → Amount of data per second

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17. What is EBS Multi-Attach?
Answer:
EBS Multi-Attach allows certain EBS volume types to be attached to multiple supported EC2 instances at the same time. The application and file system must be designed to safely handle shared access.
For example:
A clustered application designed for shared block storage may use an EBS volume with Multi-Attach.

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18. What is the difference between EBS and Instance Store?
Answer:
EBS is persistent block storage, while Instance Store provides temporary local storage attached to the physical host. Instance Store data can be lost when the instance is stopped, terminated, or the underlying host fails, depending on the situation.
For example:
EBS
→ Persistent Storage

Instance Store
→ Temporary Local Storage

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19. What happens to EBS when EC2 is stopped?
Answer:
For a normal EBS-backed EC2 instance, the EBS volume remains available when the instance is stopped. When the instance is started again, the volume can be attached and used again.
For example:
EC2 Running
↓
Stop
↓
EBS Data Remains
↓
Start EC2

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20. What happens to EBS when EC2 is terminated?
Answer:
It depends on the volume's **Delete on Termination** setting. The root EBS volume is commonly configured to be deleted when the EC2 instance is terminated, while additional data volumes may be configured to remain.
For example:
EC2 Terminated
↓
Delete on Termination = Yes
↓
EBS Deleted

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21. What is Delete on Termination?
Answer:
Delete on Termination controls whether an EBS volume should be deleted when the attached EC2 instance is terminated. We should be careful with this setting for important data volumes.
For example:
For a production data volume, we may want the volume to remain after EC2 termination.

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22. What is an EBS Volume Attachment?
Answer:
A volume attachment is the relationship between an EBS volume and an EC2 instance. Once attached, the operating system can detect the block device and mount or format it as required.
For example:
EBS Volume
↓
Attach
↓
EC2
↓
Operating System

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23. Can EBS be attached to an EC2 in another Availability Zone?
Answer:
Normally, an EBS volume is tied to a specific Availability Zone. It cannot simply be attached directly to an EC2 instance in a different AZ. We can create a snapshot and create a new volume in the required AZ.
For example:
EBS - AZ-1
↓
Snapshot
↓
New EBS - AZ-2

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24. What is the EBS Dashboard?
Answer:
The EBS resources are mainly managed through the EC2 console. From there, we can manage volumes, snapshots, volume-related settings, and other storage resources. The exact console layout can change over time.
Example:
EC2 Console
│
├— Volumes
├— Snapshots
├— Volume Types / Settings
└— EBS-related resources

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25. What is an EBS Volume Status Check?
Answer:
EBS provides status information that helps us understand whether a volume is functioning normally. This can help identify possible storage problems.
For example:
EBS Volume
↓
Status Check
↓
Healthy / Problem

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26. What are EBS Fast Snapshot Restore and FSR?
Answer:
Fast Snapshot Restore helps make volumes created from snapshots available for full performance faster in selected Availability Zones. Without it, a newly created volume from a snapshot may need to initialize data as it is accessed.
For example:
If we need to quickly create a high-performance volume from a snapshot for an important workload, FSR can help reduce the initialization delay.

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27. What is the EBS Recycle Bin?
Answer:
EBS resources such as snapshots and volumes can be protected using Recycle Bin rules. If a protected resource is deleted, it can be retained temporarily according to the configured rule instead of being immediately lost.
For example:
Delete Snapshot
↓
Recycle Bin Rule
↓
Keep Temporarily
↓
Recover if Needed

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28. How do you monitor EBS?
Answer:
EBS can be monitored using Amazon CloudWatch metrics. We can monitor storage-related performance and usage metrics and create alarms when required.
For example:
EBS
↓
CloudWatch
↓
Metrics
↓
Alarm

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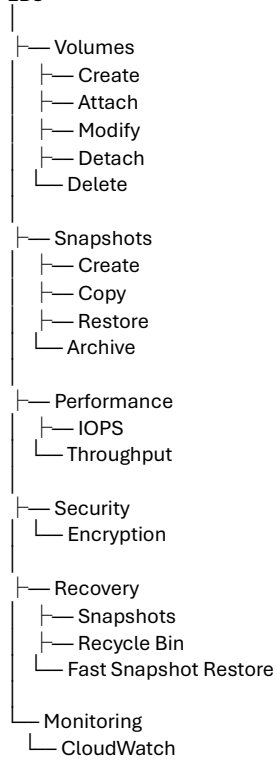
29. What is the difference between EBS Snapshot and EFS?
Answer:

An EBS Snapshot is a backup of a block-storage volume. EFS is a live shared file system that multiple EC2 instances can access. They solve different problems.
For example:
EBS Snapshot → Backup of EBS Volume

EFS → Shared File Storage

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30. EBS Console — Easy Memory Map
Answer:
Remember EBS through the EC2 console like this:



Simple ga gurthupettuko:
EBS = EC2 Block Storage + Volume + Snapshot + IOPS + Throughput + Encryption